

ORIGINAL ARTICLE

Sulfur cycling at natural hydrocarbon and sulfur seeps in Santa Paula Creek, CA

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Abstract

Biogeochemical cycling of sulfur is relatively understudied in terrestrial environments compared to marine environments. However, the comparative ease of access, observation, and sampling of terrestrial settings can expand our understanding of organisms and processes important in the modern sulfur cycle. Furthermore, these sites may allow for the discovery of useful process analogs for ancient sulfur-metabolizing microbial communities at times in Earth's past when atmospheric O₂ concentrations were lower and sulfide was more prevalent in Earth surface environments. We identified a new site at Santa Paula Creek (SPC) in Ventura County, CA—a remarkable freshwater, gravel-bedded mountain stream charged with a range of oxidized and reduced sulfur species and heavy hydrocarbons from the emergence of subsurface fluids within the underlying sulfur- and organic-rich Miocene-age Monterey Formation. SPC hosts a suite of morphologically distinct microbial biofacies that form in association with the naturally occurring hydrocarbon seeps and sulfur springs. We characterized the geology, stream geochemistry, and microbial facies and diversity of the Santa Paula Creek ecosystem. Using geochemical analyses and 16S rRNA gene sequencing, we found that SPC supports a dynamic sulfur cycle that is largely driven by sulfide-oxidizing microbial taxa, with contributions from smaller populations of sulfate-reducing and sulfur-disproportionating taxa. This preliminary characterization of SPC revealed an intriguing site in which to study geological and geochemical controls on

microbial community composition and to expand our understanding of sulfur cycling in terrestrial environments.

KEYWORDS

hydrocarbon seep, microbial mats, Sulfur cycling, sulfur isotopes

1 | INTRODUCTION

The global biogeochemical cycling of sulfur is intrinsically tied to a diverse suite of microbial metabolic reactions, which drive sulfur speciation and impart significant isotope effects (Fike et al., 2015). Supported by numerous studies on sulfur-metabolizing communities in the modern ocean water column and sediments, these geochemical signals have been applied as proxies for the presence of similar metabolisms in the geologic record, and have provided important constraints on the evolution of life and Earth's environment (Bontognali et al., 2012; Canfield & Raiswell, 1999; Farquhar et al., 2000; Fike et al., 2015; Shen et al., 2001). In contrast, the microbially mediated cycling of sulfur in terrestrial environments has received comparatively less study, though several sites have revealed rich and diverse biological-geological interactions (Baker & Banfield, 2003; Macalady et al., 2007). Typically, freshwater bodies and flows are characterized by orders of magnitude lower concentrations of sulfate than the marine realm ($[\text{SO}_4^{2-}] = 0.01\text{--}0.8\text{ mM}$, compared to seawater $[\text{SO}_4^{2-}] = 28\text{ mM}$) and are impacted by relatively steep gradients in pH, reduction potential, and temperature (Phelps & Zeikus, 1985; Capone & Kiene, 1988; Holmer & Storkholm, 2001). These distinct environmental conditions—and the heterogeneous and often time-varying sourcing of sulfur—modulate the activity of sulfur-metabolizing microorganisms and render the terrestrial cycling of sulfur particularly dynamic (e.g., Cook & Kelly, 1989; Holmer & Storkholm, 2001; Briée et al., 2007).

Freshwater bodies rich in sulfur compounds can serve as interesting field sites in which to study the impacts of water flow and sunlight on microbially mediated sulfur cycling. These ecosystems are often influenced by numerous environmental variables, which create heterogeneous geochemical niches that are occupied by morphologically, physiologically, and phylogenetically diverse microorganisms. Stratified freshwater lakes and lakes with sulfidic inputs can contain niches with varying exposure to light, oxygen, and sulfur compounds that host diverse communities of phototrophic and lithotrophic sulfur cycling bacteria involved in sulfur oxidation, reduction, and disproportionation (Block et al., 2021; Chaudhary et al., 2009; Diao et al., 2017; Hamilton et al., 2017; Holmer & Storkholm, 2001; Ingvorsen et al., 1981; Sharrar et al., 2017; Urban et al., 1994). Similarly dynamic conditions can also occur in sulfidic stream environments, where water flow provides an additional physical parameter that can influence the succession of distinct microbial communities under fluctuating environmental conditions (Klatt et al., 2016, 2020; Macalady et al., 2008). In order to better characterize the modern biogeochemical cycling of sulfur and untangle the

ecology and evolution of sulfur-metabolizing microorganisms, it is useful to develop readily accessible freshwater microbial community analogs in which a range of sulfur-based metabolic activities can be easily studied.

Here, we characterized the geology, stream geochemistry, and composition of the autochthonous microbial communities in a short (~175 m) stretch of Santa Paula Creek (SPC), in Ventura County, CA. SPC is a sunlit and oxygenated fluvial system with a well-developed systematic array of biofacies. This easily accessible site comprises a pool-and-riffle sequence charged with heavy hydrocarbons and reduced sulfur from the underlying organic-rich Miocene Monterey Formation. The significantly elevated concentrations of sulfurized organic carbon, sulfide, and sulfate in connate waters of the Monterey Formation, high rates of bedrock uplift and physical erosion, and the resulting highly concentrated, isotopically distinct sulfate reservoir in the stream (Burke et al., 2018; Petsch et al., 2000; Zaback & Pratt, 1992) make SPC a valuable field site in which to study the dynamics of terrestrial microbially mediated sulfur cycling.

Over three sampling excursions spanning 2 years, we detailed distinct microbial assemblages present in different biofacies. To investigate the microbial diversity at SPC and the communities' potential roles in sulfur redox chemistry, we gathered 16S rRNA gene amplicon data and geochemical analyses in both the oxygenated main body of the stream as well as stagnant sulfidic pools adjacent to weathering bedrock. These data revealed that primary production and sulfur cycling at SPC are primarily driven by phototrophic and chemolithotrophic sulfide-oxidizing microbial taxa. Smaller anaerobic populations within biofacies may contribute to the reductive and disproportionative branches of the microbial sulfur cycle at SPC. These communities appeared to be robust across sampling years, making the Santa Paula Creek site a useful site for future studies to investigate and isolate the impact of hydrocarbon and sulfur fluxes on terrestrial communities, and potentially serve as a process analog site for understanding the mechanics of ancient sulfur-rich microbial ecosystems.

2 | MATERIALS AND METHODS

2.1 | Geologic context of Santa Paula Creek

SPC drains approximately 114 km² as part of the larger Santa Clara River Watershed located within the Western Transverse Ranges of the central Ventura Basin, Ventura County, California. The Santa Clara River Watershed is largely mountainous and drains an area

extending from Pacific Mountain in the San Gabriel Mountain range to the Pacific Ocean just south of the City of Ventura. The total drainage area is approximately 4200 km² and represents one of the largest mostly natural river systems in Southern California (Stoecker & Kelley, 2005).

SPC is located in the Ventura Basin, which contains strata ranging from deep marine trough turbidities to shallow-marine and non-marine siliciclastic-rich strata that were deposited in mid Miocene to early Pliocene time (ca. 20–5 Ma). The SPC headwaters mainly flow through the Eocene-aged Matilija Sandstone and Coldwater Sandstone formations before dropping into Quaternary deposits and incising a plunging anticline of the Monterey Formation, Sisquoc Shale, and Pico Formations at the northeastern end of Sulphur Mountain. The geology of the SPC field site is dominated by the Sulphur Mountain anticline, which began to form as a fault-propagation fold beginning in the middle Pleistocene (ca. 2.0–2.2 Ma) and is composed of upper Miocene Monterey Formation strata in its core (Figure 1b) (Dibblee, 1990; Huftile, 1991). This formation is rich in organics and is dominated by thinly bedded diatomaceous or siliceous shale of Mohnian Stage (upper Miocene) (Dibblee, 1990). Bounding the anticline is the Sulphur Mountain reverse fault to the south and the Sisar reverse fault to the north (Huftile, 1991). SPC runs through the Quaternary fan and terrace deposits upstream of the sample site before it cuts across the eroded crest of the Sulphur Mountain anticline crossing the Monterey Formation. The Monterey Formation has been fractured by the nearby Sisar Fault; with no effective oil trapping mechanism, natural oil and sulfur seeps updip from the strata into SPC (Huftile, 1991). The bedrock contains a deep weathering profile beneath the overlying Pleistocene depositional (strath) terrace. Tar seeps and sulfur springs are predominantly

found on, or adjacent to faults, which act as conduits for the tar and sulfur-rich spring waters.

2.2 | Site mapping and sample collection

The SPC stream reach we mapped and sampled was approximately 175 m and runs roughly parallel to California State Route 150. The sample site was accessed via a foot trail directly opposite a small parking area off SR 150 (34°25′35.1″N 119°05′21.3″W) that provided a steep route down to the stream bed. The uppermost end of the sample site is approximately 100 meters downstream from the bridge to Thomas Aquinas College. A foot trail starting on the northeast side of the road immediately after the bridge provided alternative site access.

At the time of sampling in June of 2017, September of 2017, and June of 2018, SPC was approximately half bankfull with a mean depth of ~0.5 m and width of ~3 m. The creek streambed was composed of a combination of alluvial gravel, (pebbles, cobbles, and boulders with a D₅₀ of ~5 cm), and rare patches of exposed bedrock. The creek developed classical undulating pool and riffle bedforms along the bed, characteristic of its high mountain slopes. The geometry of the pool and riffle system was mapped and recorded in both years by measuring lengths and azimuthal alignment of the riffles and approximating each pool as an ellipse (Figure 1c). Lengths were measured using a tape and laser range finder while orientation was noted and recorded with a compass for each feature.

Numerous hydrocarbon seeps exist throughout the SPC site, typically emanating from cracks in the rocks forming the creek banks. Fresh seep material is best described as a moderately viscous

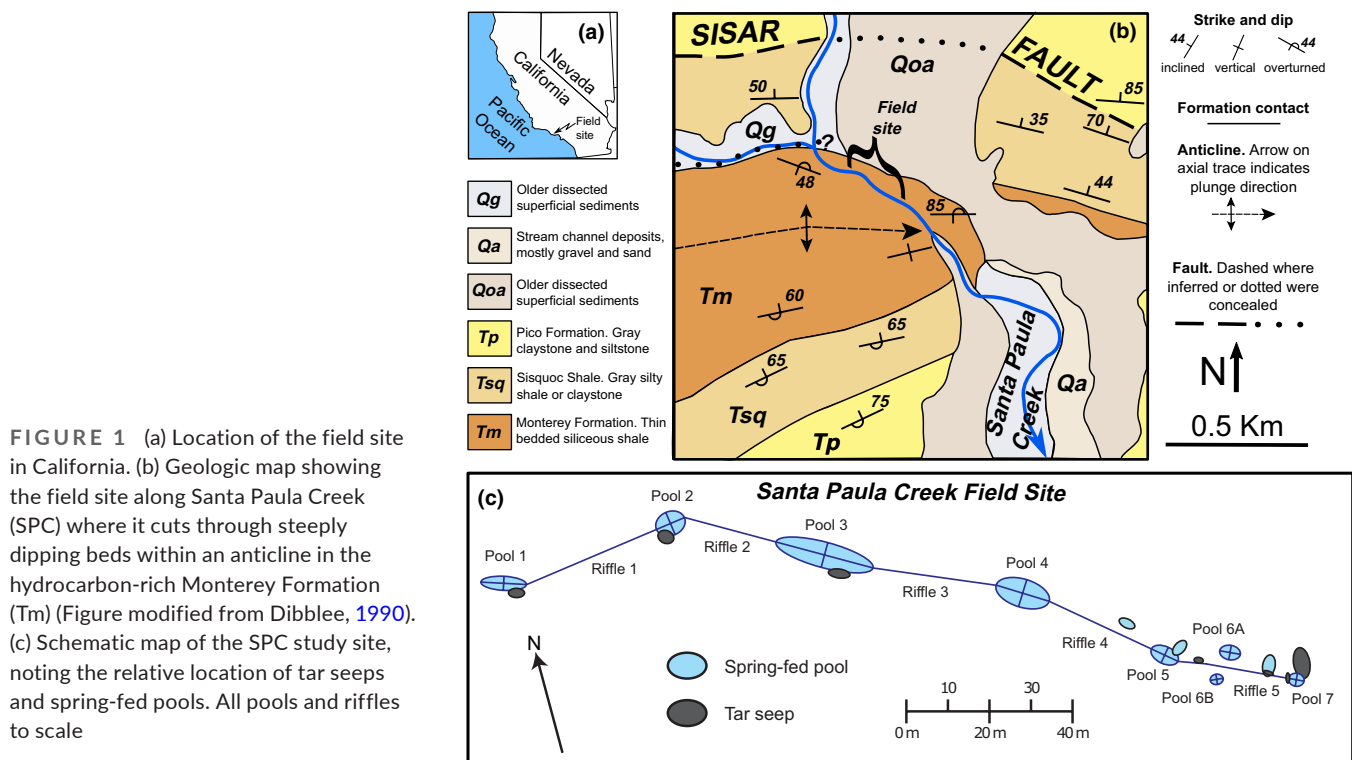


FIGURE 1 (a) Location of the field site in California. (b) Geologic map showing the field site along Santa Paula Creek (SPC) where it cuts through steeply dipping beds within an anticline in the hydrocarbon-rich Monterey Formation (Tm) (Figure modified from Dibblee, 1990). (c) Schematic map of the SPC study site, noting the relative location of tar seeps and spring-fed pools. All pools and riffles to scale

(~pancake syrup), sulfur-rich, heavy oil typical of those often sourced by thermally immature rocks of the Monterey formation. It flows slowly downhill into the creek, where it floats. Where these seeps encounter moving water, the oil is rapidly entrained and moved downstream where it sticks to rocks or the legs of unsuspecting scientists, however when they enter quiet pools, they can form a substantial (up to 1 cm thick) floating "lid" of oil that greatly limits gas exchange and causes the underlying waters to become O_2 -depleted or even anoxic. With progressive degradation, oxidation, and volatile loss the heavy oil becomes more and more viscous, eventually reaching a solid state that is akin to asphalt and can be walked on. The full range of physical compositions is present in SPC, depending on the residence time of oil in the local environment. For simplicity, we refer collectively to all such materials as bitumen.

The presence of bitumen, sulfur seeps, sediment, and microbial biofacies was mapped in each designated riffle and pool and these sites were selectively targeted for sampling. In June 2017 and 2017, biomass samples were collected aseptically and placed in either 15 ml falcon tubes containing RNeasy lysis solution for subsequent DNA extraction and 16S rRNA gene amplicon sequencing, or in microcentrifuge tubes containing 4% paraformaldehyde for microscopy. Red-green biofilms ($n = 14$), green-purple mats ($n = 6$), dark green mats ($n = 4$), and yellow-green mats ($n = 9$) were collected from stagnant and slowly flowing pools at SPC. Filamentous streamers (dark green, $n = 4$; mint green, $n = 14$; white, $n = 10$) were collected from turbulent riffles. Sediment ($n = 12$) and bitumen ($n = 5$) samples were also collected aseptically and preserved for DNA extraction and 16S rRNA gene amplicon sequencing. All biomass samples were described and photographed prior to preservation. In June 2017, macrofauna were collected (midge larvae, $n = 2$; snails, $n = 1$) or swabbed (tadpoles, $n = 2$; frogs, $n = 2$) for DNA extraction and 16S rRNA gene amplicon sequencing of their associated microbiomes. Biomass samples were not collected in September 2017. Samples were collected for geochemical analyses during all three sampling periods and are described in the following section.

2.3 | Geochemical analyses

Samples for geochemical analyses were collected during three sampling trips: June 2017, September 2017, and June 2018 (Table 1). In situ measurements of dissolved oxygen, pH, and temperature were made in June 2017 and 2018 sampling trips using a portable pH/mV/Temp probe (Extech DO700 8-in-1 Portable Oxygen Meter; FLIR Commercial Systems, Inc., Nashua, NH) that was calibrated in the field prior to use.

In June 2017, fluid samples were collected for measurement of sulfate concentrations and isotope ratios of specific sulfur species; fluid samples were reacted with 0.5 M zinc acetate to capture dissolved sulfide, and those samples were wrapped to prevent photochemical oxidation. After samples were returned to the lab, the precipitated zinc sulfide was then centrifuged, rinsed three times with MilliQ water, and dried before analysis. Samples for sulfide

isotopic composition were analyzed on an EA-IRMS for $\delta^{34}S$ (a Thermo Fisher Scientific Flash Elemental Analyzer [EA] Isolink connected to a Delta V Plus Isotope Ratio Mass Spectrometer [IRMS] via a ConFlo IV Universal Interface, or EA-IRMS). Sulfate samples were analyzed using a Dionex Ion Chromatography System 2000 and a Neptune™ Multi-Collector-Inductively Coupled Plasma-Mass Spectrometer (Thermo Scientific Neptune+ multicollector ICP-MS) at Caltech for their concentrations and $\delta^{34}S$ following the method in (Paris et al., 2013).

In September 2017, samples were collected for measurements of sulfate concentration, isotopic composition of sulfate and sulfide, nitrate/nitrite concentration and isotopic composition, and concentration of total dissolved nitrogen (TDN). Samples for sulfide measurements were collected by filtering an aqueous sample directly into a falcon tube via a syringe fitted with a 0.2 μm filter and needle containing a pre-weighed volume of 0.5 M zinc acetate in order to immediately precipitate any sulfide. After the samples were returned to the lab, the precipitated zinc sulfide was then centrifuged and rinsed three times with MilliQ water, and dried before analyses. The supernatant of the zinc-preserved samples was used for sulfate isotopic analyses and concentration measurements. The isotopic compositions ($\delta^{34}S$) of sulfide and sulfate were both measured on an EA-IRMS (described above). Sulfate concentrations were measured with a Dionex Ion Chromatography System 2000. Sulfate concentration and $\delta^{34}S$ have estimated uncertainties based on standard replicates of $\pm 3\%$ and $\pm 0.3\%$, respectively.

Stream water for nitrogen analyses was collected and 0.2 μm syringe filtered in the field. Nitrate concentrations were analyzed using the chemiluminescence method (Braman & Hendrix, 1989). The isotopic composition ($\delta^{15}N$) of nitrate was analyzed using the denitrifier method (Sigman et al., 2001). Note that these methods do not distinguish between nitrate and nitrite. Hereafter, nitrate refers to nitrate plus nitrite. TDN was analyzed by persulfate oxidation of TDN into nitrate, followed by measurements of the resulting nitrate for its concentration and isotopic composition using the denitrifier method (Wang et al., 2015). The analytical precisions for the concentration and $\delta^{15}N$ are $\pm 3\%$ and $\pm 0.2\%$, respectively.

In June 2018, samples were collected for sulfide concentration and bulk sulfur $\delta^{34}S$ values within filamentous streamers. Water samples collected from the stream and springs were syringe filtered in the field and carefully injected into a pre-weighed glass bottle of 0.5 M zinc acetate in order to minimize oxidation and loss of sulfide. The samples were then centrifuged in the lab. The supernatant was collected for sulfate analyses. The precipitate was rinsed with MilliQ water three times. The rinsed precipitate was pipetted sulfide concentration measurements via the Cline assay by reacting water samples with the methylene blue reagent and measuring absorbance at 670 nm on a spectrophotometer at Caltech (Cline, 1969). White streamer bulk sulfur $\delta^{34}S$ values were measured by EA-IRMS.

In June 2018, silver strips were immersed in green-purple microbial mats under daylight and "night" conditions, which were simulated by covering the mat with an opaque foil for 20 min. Sulfide in the mat reacted with the silver to produce Ag_2S , which

TABLE 1 Summary of geochemical analyses performed on SPC samples

Date	[SO ₄ ²⁻]	δ ³⁴ S SO ₄ ²⁻	[H ₂ S]	δ ³⁴ S H ₂ S	[NO ₃] ^a	TDN	δ ¹⁵ N
June 2017	MC-ICP-MS	MC-ICP-MS	NA	EA-IRMS	NA	NA	NA
Sept. 2017	EA-IRMS	EA-IRMS	NA	EA-IRMS	Chemi. ^b	Chemi.	EA-IRMS
June 2018	NA	EA-IRMS (streamers only)	Cline ^c	NA	NA	NA	NA

^aMeasured nitrate concentration is a total of nitrate and nitrite concentrations.

^bChemiluminescence method (Braman & Hendrix, 1989).

^cCline (methylene blue) assay (Cline, 1969).

accumulated on the strip (Fike et al., 2008, 2009). Approximately 1 cm of the strip was exposed to the overlying water, and the water-mat interface was marked following incubation using a permanent marker. Sulfur abundance and δ³⁴S of the sulfides deposited on the silver strips were analyzed using a Cameca IMS 7f-GEO magnetic sector secondary ion mass spectrometer (SIMS). Silver strips were optically imaged using a flatbed scanner and then cut into approximately 1-inch sections to fit into the SIMS sample holder. The SIMS primary beam current was ~3 nA, rastered over an area of 200 μm × 200 μm. Secondary ions were collected using Faraday cups. Automatic peak centering and mass calibration were performed at the start of each measurement to account for possible magnetic field or secondary ion beam position drift. Samples were pre-sputtered for 30 s with the same beam current. Measurements were collected in cycles as described by Fike and coworkers (Fike et al., 2009), and internal error was typically <1%. Sample analyses were bracketed by analyses of a standard sample, which was a similar silver strip exposed to a sodium sulfide solution of known isotopic composition (δ³⁴S = 16‰). All sulfur isotope values are reported relative to the V-CDT (Vienna Canyon Diablo Troilite) scale. Relative sulfur counts were calculated by normalizing absolute counts to the minimum and maximum values for the entire dataset.

2.4 | Sample imaging: light microscopy, SEM, and XANES

Samples were analyzed via scanning electron microscopy (SEM) on a ZEISS 1550 VP field emission SEM. PFA-preserved samples were deposited on polycarbonate filters (GTTP Isopore membrane filters, Merck Millipore, pore size 0.2 μm) and gently rinsed with distilled water. The filters were dried and coated with platinum prior to analysis. Images were acquired using primarily in-lens SE and SE2 detectors, applying angle-sensitive backscatter (AsB) mode to highlight sulfur granules with a 10–15 kV electron beam for imaging and a working distance of ~9 mm. Samples were assessed for elemental distribution and mapped using an Oxford X-Max SDD Energy Dispersive X-Ray Spectrometer (EDS) system (relative accuracy of <5% and detection limit of <0.5%), with a 15 kV electron beam. Samples were also imaged using a standard light compound microscope with samples prepared on wet mounts and images were captured using an eyepiece camera.

SPC samples were analyzed and imaged via X-ray absorption near edge structure (XANES) spectroscopy at the sulfur K-edge using the X-ray microprobe beamline 14-3 at the Stanford Synchrotron Radiation Lightsource (SSRL) at SLAC National Accelerator Laboratory, Menlo Park, CA. Streamer samples collected and stored in 15 ml falcon tubes with no preservative were gently rinsed with DI water, placed on a 10.0 μm polycarbonate membrane filter (Type TCTP, Millipore), and allowed to air dry. Asphalt samples were prepared by placing drops of different asphalt samples directly onto a silicon chip (Active Business Company GmbH, Riederling, Germany). The filters or chips were then taped to a sample mount in a helium environment (purged to <1% oxygen) on beamline 14-3 for analysis. Energy calibration was done using an inline thiosulfate standard. Samples were analyzed for sulfur K-edge measurements and an initial coarse map of total sulfur was generated. To find analytical regions of interest the asphalt samples were first imaged at an energy of 2500 eV, dwell time of 10 ms, and step distance of 10 μm, while filament samples were measured with an energy of 2500 eV, dwell time 25 ms, step distance of 25 μm. Smaller areas of interest were selected on each asphalt sample for a fine energy spectral scan, spanning an energy range of 2400–2500 eV, dwell time of 10 ms, and beam size of ~4 μm. In filament samples, smaller areas of interest were selected on each sample for a fine energy sulfur map using the following energies to help distinguish key sulfur moieties in the samples (2472.6 eV – elemental sulfur, 2473.6 eV – methionine, 2476.3 eV – sulfoxide, 2481.2 eV – sulfonate, 2500 eV – total sulfur) with a dwell time of 25 ms and step distance of 4.0 μm.

2.5 | 16S rRNA gene amplicon sequencing and analysis

In 2017 and 2018, DNA was extracted from a total of 77 samples preserved in RNA_{later} using the Qiagen PowerSoil DNA extraction kit with the following modifications: samples were bead beaten for 45 seconds at 5.5 ms⁻¹ and 75 μl solution C6 was added to the filter membrane in the spin column and allowed to rest for 1–2 min (Table S1). All extractions were performed on the same day and one extraction blank was performed per extraction kit of 50 samples. The DNA yield of the extracted samples and blanks was verified via gel electrophoresis. The V4-V5 hypervariable region of the 16S rRNA gene was amplified by PCR using universal primer set 515F and 926R modified with Illumina adapters on 5' end (515F 5'

-TCGTCGGCAGCGTCAGATGTGTATAAGAGACAG-GTGYCAGC MGCCGCGGTAA-3' and 926R 5'-GTCTCGTGGGCTCGGAGATG TGTATAAGAGACAG-CCGYCAATTMTTTRAGTTT-3'). The V4 hypervariable region of the 16S rRNA gene from the June 2017 animal microbiome samples from ($n = 8$) was amplified using the 515F and 806R primers modified with Illumina adapters on the 5' end (515F 5'- AATGATACGGCGACCACCGAGATCTACACGCT-GTGCCAGCMGCCGCGGTAA-3' and 806R 5'- CAAGCAGAAGA CGGCATACGAGAT- GGACTACNVGGGTWTCTAAT-3') (Caporaso et al., 2011). Duplicate PCR reactions were set up for each sample with Q5 Hot Start High-Fidelity 2x Master Mix (New England Biolabs) in a 15 μ l reaction volume, with annealing at 54°C, for 28 cycles. Amplification of the 16S rRNA gene was verified via gel electrophoresis. Duplicate PCR samples were then pooled and barcoded with Illumina Nextera XT index 2 primers that include unique 8-bp barcodes (P5 5'-AATGATACGGCGACCACCGAGATCTACAC-XXXX XXXX-TCGTCGGCAGCGTC-3' and P7 5'-CAAGCAGAAGACGGCA TACGAGAT-XXXXXXXXX-GTCTCGTGGGCTCGG-3'). Amplification with barcoded primers used Q5 Hot Start PCR mixture but used 2.5 μ l of product in 25 μ l of total reaction volume, annealed at 66°C, and cycled 10 times. Products were purified using Millipore-Sigma MultiScreen Plate MSNU03010 with vacuum manifold and quantified using Thermo Fisher Scientific QuantIT PicoGreen dsDNA Assay Kit P11496 on the BioRad CFX96 Touch Real-Time PCR Detection System. Barcoded samples were combined in equimolar amounts into single tubes and purified with Qiagen PCR Purification Kit 28,104 before sequencing on Illumina MiSeq with the addition of 15%–20% PhiX and with a 2x250 protocol (Laragen Inc.). 16S rRNA amplicon sequencing data are available in GenBank under BioProject accession number PRJNA841681.

Demultiplexed MiSeq 16S rRNA data from 2017 and 2018 were processed together QIIME2 v2019.7.0 (Bolyen et al., 2019). Sequences were imported into QIIME2 using the FastQ manifest format, demultiplexed and denoised into amplicon sequence variants (ASVs) using the QIIME2 DADA2 denoised-paired plugin (Callahan et al., 2016). Taxonomic identification for representative sequences of each ASV was assigned using the SILVA 138 database at 99% sequence similarity using the feature-classifier classify-sklearn plugin using a classifier trained on the SILVA138 database clustered at 99% and trimmed to the amplified region. (Quast et al., 2012). Sequences assigned to mitochondria and chloroplasts (7.7% of total sequences) were removed using the taxa filter-table plugin. The filtered OTU table and taxonomy were used for further analyses and visualization of data using the R package phyloseq (McMurdie & Holmes, 2013). Sequences that were unassigned at the domain level were removed and sequence counts were transformed to relative abundance. Interpretations of microbial metabolisms at SPC were inferred from the metabolisms of the closest cultured relatives. Statistical analyses, including diversity indices and non-metric multidimensional scaling analyses, were conducted and plots were produced in R using the R packages vegan and ggplot2 (Figures S4 and S5) (Oksanen et al., 2019; R Core Team, 2019; Wickham, 2016). Alpha diversity was estimated using the Shannon index and Simpson

index (Shannon, 1948; Simpson, 1949) and significance levels were calculated using the *t*-test.

3 | RESULTS

3.1 | Site map and biofacies description

At the study site, SPC is a mountain stream exhibiting bedforms of a classic a pool and riffle sequence alternating between shallow, turbulent gravel-bed riffles and deeper pools. Overhanging vegetation at the creek included bushy chaparral and large live oaks. At the time of sampling, stream discharge measured at a USGS gauge (site 11,113,500, coordinates = 34.413333, -119.081389) approximately 1 km downstream from the study area was $0.20 \text{ m}^3 \text{ s}^{-1}$ and $0.14 \text{ m}^3 \text{ s}^{-1}$ in 2017 and 2018, respectively—amounts similar to the long-term 50% exceedance value for this location. During these times the channel was filled to a reach-integrated mean depth of ~0.5 m and width of ~3 m. The sampling transect featured numerous proximal bitumen seeps (Figure 2a) and spring-fed pools (Figure 2b) that discharge into SPC (Figure 1c). Site maps generated each year were similar from 2017 and 2018, with several additional pools noted in 2018, likely related to a reduced streamflow that year and limited local, intermittent sediment transport associated with high flows during winter storms. The unique microbial biofacies present in different pools (described below) were generally consistent between years, suggesting that the biogeochemical conditions supporting them are largely steady from year-to-year. However, this region of Southern California undergoes episodic wet years largely driven by El Niño Southern Oscillation, which can affect biofacies persistence due to elevated discharge and sediment transport following a wetter winter. For example, an El Niño flood in 2007 completely scoured the streambed and eroded several meters downward. The streamers, microbial mats, and biofilms had returned by the following year. In January of 2019, large precipitation events during winter storms that year yielded peak discharge in SPC of $18.0 \text{ m}^3 \text{ s}^{-1}$. A subsequent site visit in April of that year revealed massive sediment transport (up to m-scale boulders) that scoured nearly all of the macroscopic biofacies from the bed. However, by the end of summer, we observed that these biofacies had recovered.

Two distinct morphological microbial biofacies (biofilms/microbial mats and filamentous streamers) were consistently observed within SPC during the 2017 and 2018 sampling (Figure 2). The two morphologies were present in distinct regions of the creek based on apparent differences in water flow. Biofilms and microbial mats were observed in more stagnant or slower-flowing pools, often attached to the banks of the stream. Biofilms were composed of thin, red-green laminae on the creek sediment (Figure 2c) where they were exposed to direct sunlight. Mats were thicker (~10 cm) and composed of alternating laminae of green-purple (Figure 2d), yellow-green (Figure 2e), or dark green color (Figure 2f), and occurred in directly sunlit and shaded regions of the stream. Filamentous streamers, consisting of filament bundles approximately 1–3 m long and several cm thick,

FIGURE 2 Bitumen, sulfur springs, and biofacies at Santa Paula Creek. (a) Bitumen seep. (b) Sulfur spring water. (c) Red-green biofilm. (d) Purple-green microbial mat. (e) Yellow-green microbial mat. (f) Dark green microbial mat. (g) White filamentous streamer. (h) Mint green filamentous streamer. (i) Dark green filamentous streamer



were observed anchored to rocks or sediment within fast flowing, turbulent riffles in the stream. The streamers ranged in color from white (Figure 2g), to mint green (Figure 2h), to dark green (Figure 2i). Samples were also collected from bitumen seeps (Figure 2a) and stream sediments. Active bitumen seeps were composed of black, viscous liquid asphalt. In addition to forming pools on the creek bank, oil was observed seeping directly into the stream. Balls of bitumen 1–5 cm in diameter were observed floating atop the water in riffles. In more stagnant pools or near active bitumen seeps, a coating of oil coagulated at the water surface. Sediment was gray or black.

Biofacies contained diverse assemblages of microeukaryotes that were observed by light and electron microscopy (Figure 3). Streamers contained persistent structures that consisted of a central green algal filament (*Oedogonium*, *Vaucheria*, *Ulothrix*, *Tribonema*, and *Microspora* as identified by light microscopy) coated with pennate and gomphonemate diatoms, connected and bound by bacterial filaments that contained conspicuous sulfur globules (Figure 3g,i). The waters and surface sediments of SPC hosted a typical freshwater assemblage of diatom species including *Cyclotella*, *Stephanodiscus*,

Ulnaria, and *Thlassiothrix*. The surface and upper green layers of microbial mats were dominated by pennate forms *Diatona* sp., and *Navicula*. The abundance and diversity of diatoms appeared to decrease rapidly below the green mat layers. The green upper layers of the microbial mats also contained a framework of *Oedogonium* and *Vaucheria*, with a lower abundance of other green algal taxa (see above), including free-living spherical individuals. Macro-organisms observed at SPC included frogs and tadpoles as well as various insects such as midges and dragonflies.

3.2 | Creek geochemistry

Measurements of pH, dissolved oxygen (DO), and temperature in 2017 and 2018 revealed values typical of freshwater streams in this area (Figure S1) (Reichard et al., 1999). Temperatures ranged from ~20 to 30°C, depending on the depth of measurement, local water flow rate, and level of shade over the pools. In 2017 a consistent downstream trend of decreasing pH from 7.5 to 6.5 was observed

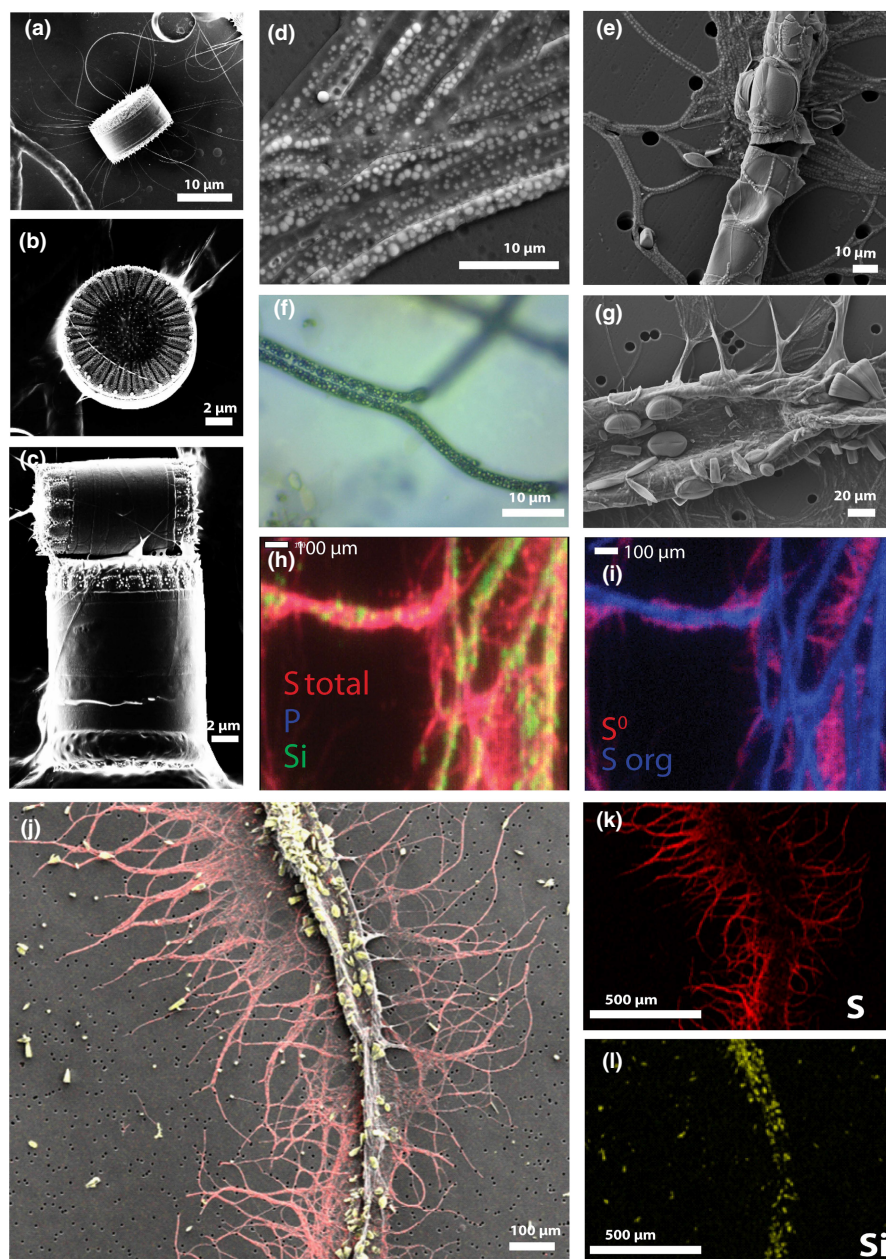


FIGURE 3 A selection of optical, SEM-EDS and XANES images of micro eukaryotes and sulfur-oxidizing bacterial associations from Santa Paula Creek. (a) *Stephanodiscus* sp., diatom. (b) *Stephanodiscus* sp., diatom ventral view. (c) *Cyclotella* sp., diatoms in different growth stages. (d) Elemental sulfur globules (white spots) within *Thiothrix* cells. (e) *Oedogonium* sp. showing the associated pennate diatoms and *Thiothrix* cells (thin threads). (f) *Thiothrix* in optical with visible elemental sulfur globules. (g) *Vaucheria* sp., algal filament with associated pennate *Frustulia* sp., (far left) and *Licomophora* sp., (far right) diatoms. *Cymbella* sp., and *Closterium* sp., diatoms are also present (along lower part of the algal filament). (h) XANES map of an algal filament indicating the presence of total sulfur (S), phosphorus (P), and silicon (Si). Light purple areas indicate P associations with S. (i) XANES map of the same filament showing the presence of organic S and elemental S. The elemental S is clearly associated with the bacterial sulfur sheaths (light purple threads emanating from the blue algal filaments). (j) Composite SEM-EDS image showing that sulfur (red) is associated with the bacterial sheaths and Si (yellow) is associated with the diatoms attached to an algal filament. (k) and (l) are the separate respective element maps of (j)

over the sampling transect, but was not noted in 2018. Similarly, DO values decreased downstream in 2017 from ~200 to ~100 μM , consistent with an O_2 demand in this reach of SPC induced by the bitumen and sulfur seeps. DO values measured in 2018 ranged between 100–350 μM depending on pool with no systematic downstream pattern. This variability suggested that the bulk stream composition may be influenced by weather, stream flow, and other hydrological processes, in addition to the biogeochemical processes occurring within the stream. One general trend noted between sampling years was that DO concentrations were most variable within the more turbulent riffles, highest when sampled above phototrophic mats, and lowest within more stagnant, sulfidic pools.

Multiple hydrocarbon seeps and sulfur springs were present along the length of the sampling transect (Figure 1c and 2a,b). The sulfur species present in Santa Paula Creek were interpreted

to derive predominantly from the seepage of formation waters that have interacted with the Monterey Formation organic- and sulfur-rich mudrocks and the subsequent petroleum system developed upon burial, deformation, and exhumation of the underlying bedrock—introduced either as sulfate through oxidation of the abundant sulfur-rich organic matter and pyrite (Zaback & Pratt, 1992) or as sulfide as observed in the anoxic springs flowing directly into SPC from the sides of the stream. The petroleum beneath Sulphur Mountain and around Santa Paula Creek was important to the indigenous Chumash people; later, these deposits were among the first commercially successful California oil fields (Hudson & Blackburn, 1987; Taylor & Welty, 1950). Samples of fresh and partially degraded asphalt were collected around 7 m downstream from the start of the study area and analyzed by sulfur K-edge XANES spectroscopy in order to determine if different organic-sulfur compounds could be

identified across a degradation gradient (Figure 4). The sample included five separate areas of analysis, including two spectra on an area that appeared freshly exposed versus three areas that appeared to be degraded based on color and texture of the analysis area. Here, “degraded” refers to oxidation and loss of volatiles in air after emergence from the seep along Santa Paula Creek, not to degradation within the reservoir. Crawford (Crawford, 1896) describes some of the petroleum from Sulphur Mountain as green in color, consistent with our observations of some samples collected from along Santa Paula Creek.

Sulfur K-edge XANES spectra were largely consistent with those previously reported for Monterey Formation asphalts (George & Gorbaty, 1989; Waldo et al., 1991). The spectra had two major peaks, at 2473.8 eV and at 2475.8 eV, which we assign to thiophene and sulfide species, respectively, based on standard spectra. In the “fresh” samples, a shoulder at ~2473 eV was appeared, consistent with the presence of reduced sulfur species not observed in the presumably more oxidized “degraded” samples. In the degraded samples, a new peak attributable to sulfate was observed at ~2482 eV. The reduced sulfur species in the fresh samples are likely mostly sulfides but may include contributions from thiols, elemental sulfur, or polyatomic sulfur cations. The green color we and others observe may be attributable to polyatomic cations of sulfur (Gillespie et al., 1971).

Dissolved sulfate concentrations collected from the SPC pool and riffle system in 2017 were significantly elevated throughout the sampled section compared to typical freshwater streams (Burke et al., 2018), with an average concentration of 1600 μM and no evidence of a downstream trend (Figure 5a). The upstream $\delta^{34}\text{S}_{\text{sulfate}}$ values were notably low and consistently clustered around -10‰ (V-CDT), whereas further downstream we observed a greater variability and higher on average $\delta^{34}\text{S}_{\text{sulfate}}$ values. We obtained a sample

of creek-adjacent spring water trickling from the black shale bedrock (outliers at ~140m; Figure 4a) that displayed a lower sulfate concentration (~600 μM) and higher $\delta^{34}\text{S}_{\text{sulfate}}$ value (+25‰). The concentration of sulfide in the main flow of SPC (measured June 2018) was remarkably high for an oxygenated stream, ranging from 5 to 15 μM with no apparent downstream trend (Figure 5b). Spring waters and bitumen seep pools had significantly elevated concentrations of sulfide, in excess of 450 μM (outlier arrows in Figure 4b). $\delta^{34}\text{S}_{\text{sulfide}}$, but not sulfide concentration, was measured in 2017. These $\delta^{34}\text{S}_{\text{sulfide}}$ values ranged between -13 to $+2\text{‰}$ and displayed a systematic pattern of increasing ^{34}S -enrichment downstream. Similar to $\delta^{34}\text{S}_{\text{sulfate}}$ data (Figure 5a), stream-adjacent spring waters had relatively higher $\delta^{34}\text{S}_{\text{sulfide}}$ values (+4.6‰). Finally, total sulfur $\delta^{34}\text{S}_{\text{org}}$ values were measured from five white streamer bulk samples along the sampling transect (Figure 5c). These data were the most ^{34}S -depleted in the dataset (-25‰ to -12‰), and similarly exhibited progressively higher values downstream.

Concentrations and $\delta^{15}\text{N}$ isotopic signatures of nitrate showed an inverse relationship along the sampling transect (Figure 5d,e). Total dissolved nitrogen (TDN) generally increased in both concentration (17–24 μM) and isotopic composition (-5.2‰ to 7.4‰ vs. air) along the downstream transect (Figure 5D). Conversely, nitrate concentrations decrease steadily from ~4.7 to ~3.3 μM , while $\delta^{15}\text{N}_{\text{nitrate}}$ increased from 6.9‰ to 8.1‰ (Figure 5e). The adjacent spring waters, which seep into the creek were sampled and found to have a nitrate concentration of 0.12 μM and a TDN concentration of 178 μM , indicating that the downstream trend in stream TDN is largely driven by dilution from spring water inputs. Although ammonium concentration was not measured directly, TDN in the spring water is likely composed of mostly ammonium, because (1) DON concentration tends to be invariant in aquatic systems due to its chemical

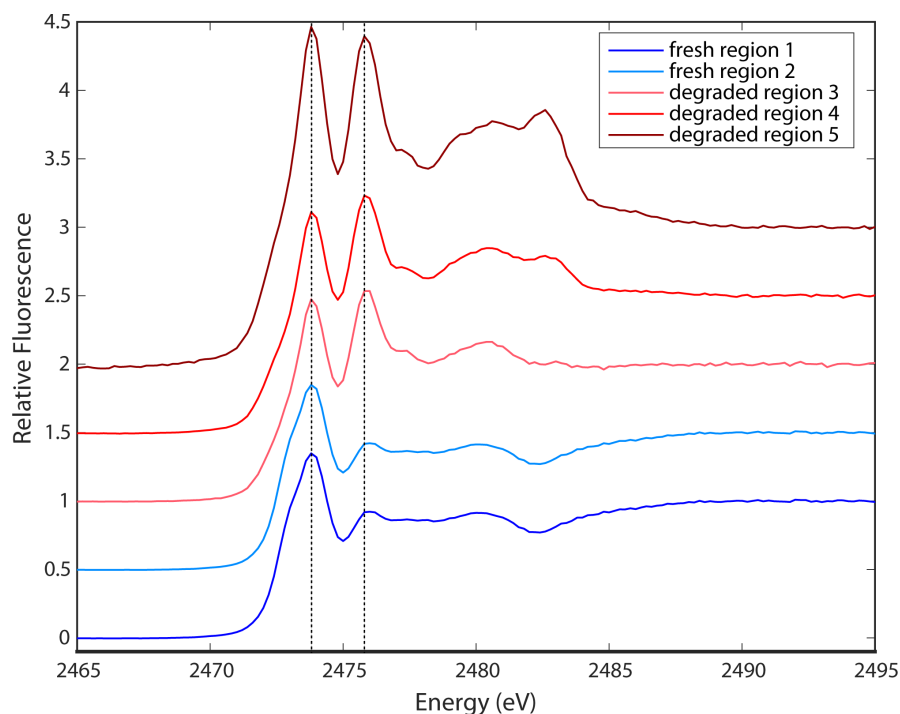


FIGURE 4 K-edge XANES spectra of a Santa Paula Creek tar sample. The spectra are stacked for clarity and include relative fluorescence. Vertical dotted lines were added to identify peaks of interest (2473.8 eV – thiophene, and 2475.8 eV – sulfide) in the sample

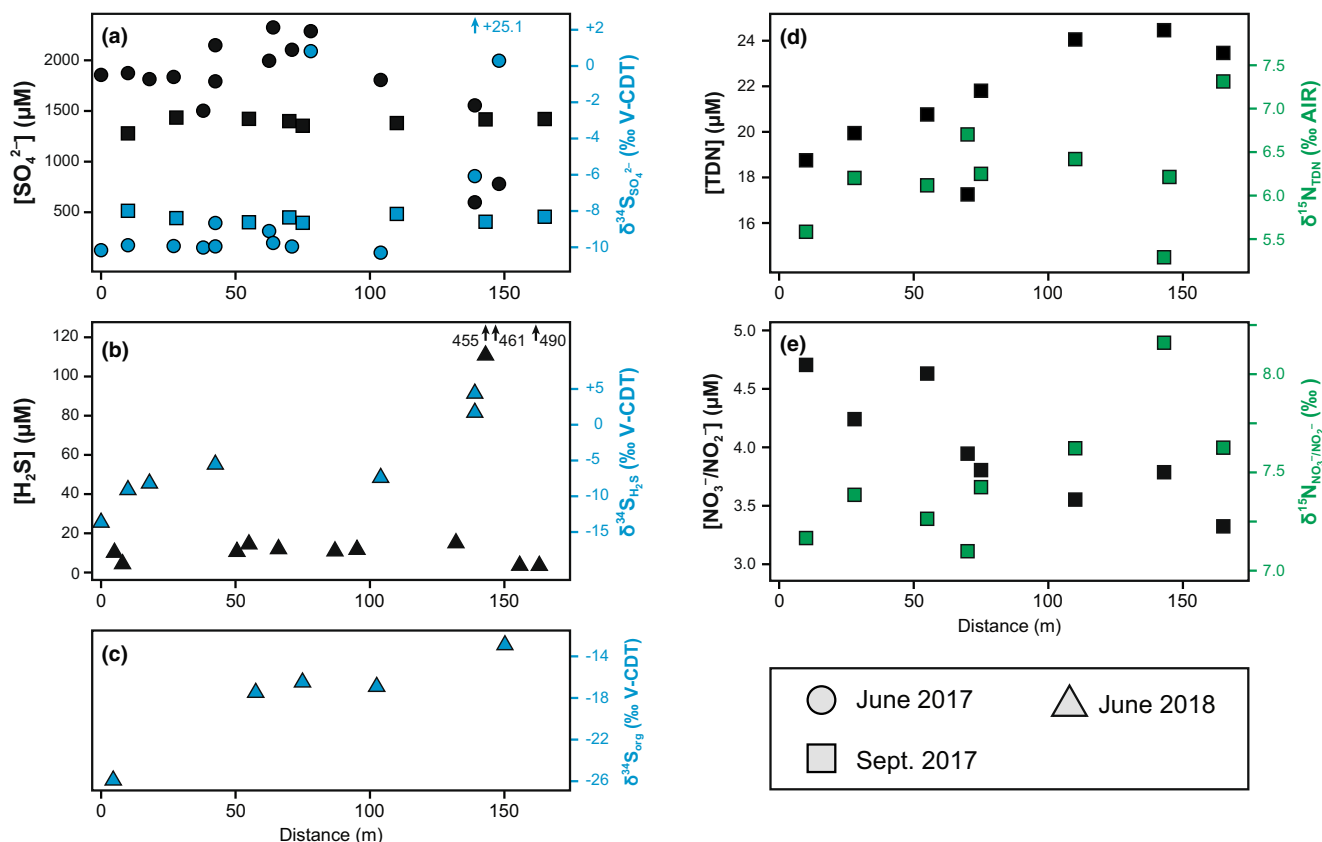


FIGURE 5 Bulk stream sulfur and nitrogen chemistry along the Santa Paula Creek transect. All black points represent concentrations, and colored points represent isotopic compositions. (a) Dissolved sulfate concentration and $\delta^{34}\text{S}$ (‰ V-CDT). Outlier $\delta^{34}\text{S}$ value (arrow) sampled from spring seep out of bedrock. (b) Dissolved sulfide concentration and $\delta^{34}\text{S}$. Outlier sulfide values (arrows) sampled from tar pools and spring seep out of bedrock. (c) Total sulfur isotopic composition sampled from white streamer biofacies. (d) Total dissolved nitrogen (TDN) concentration and $\delta^{15}\text{N}$ (‰ AIR). (e) Dissolved nitrate, measured as total nitrate plus nitrite, and $\delta^{15}\text{N}$. Point shapes correspond to the different sampling trips, as noted in legend

recalcitrance and (2) since the spring water has $>460\mu\text{M}$ sulfide, any nitrogen in the spring water is expected to be the reduced species. (i.e., ammonium).

3.3 | Microbial community composition and isotopic analyses of sulfur compounds

Community analysis by 16S rRNA sequencing was used to investigate microbial diversity and potential roles in sulfur cycling in 77 samples collected systematically from these biofacies, including from biofilms ($n = 13$), microbial mats ($n = 19$), filamentous streamers ($n = 28$), stream sediment ($n = 12$), and bitumen ($n = 5$). Taxonomy of the microbial communities was identified at the genus level, and interpretations of potential metabolisms were made based on the metabolisms of their closest cultured relatives. The Shannon and Simpson diversity indices were used to calculate species diversity within SPC biofacies communities (Shannon, 1948; Simpson, 1949). These diversity indices take into account both species abundance and evenness within a community. Streamer communities were the least diverse of all the biofacies and were significantly less diverse

($p < .001$) than the communities in all other biofacies except for bitumen (Figure S4). Biofilm communities were significantly more diverse than mat communities ($p < .05$) as calculated by the Shannon index, but not with the Simpson index.

White ($n = 10$), mint green ($n = 14$), and dark green ($n = 4$) filamentous streamers collected from riffles were colonized primarily by the filamentous sulfide-oxidizing bacterium *Thiothrix* at up to 78% relative abundance, with other genera of sulfide-oxidizing and hydrogen-oxidizing bacteria present at lower abundances (Figure 6, Figure S2). Analysis of SPC streamers by scanning electron microscopy and elemental analysis revealed silicon-rich septate central stalks (Figure 3e) with closely associated sulfur-rich filaments (Figure 3j). The sulfur-rich filaments contained numerous internal globules, visible by both light and scanning electron microscopy (Figure 3d,f). Synchrotron XANES confirmed that the sulfur in the filaments was S^0 , which was likely a product of microbial sulfide oxidation (Figure 3h,i) (Shively, 2006).

Red-green biofilms ($n = 13$), green-purple mats ($n = 6$), dark green mats ($n = 4$), and yellow-green mats ($n = 9$) were collected from more stagnant and slowly flowing pools at SPC. Microbial communities within these samples were colonized by microbes typical of mats

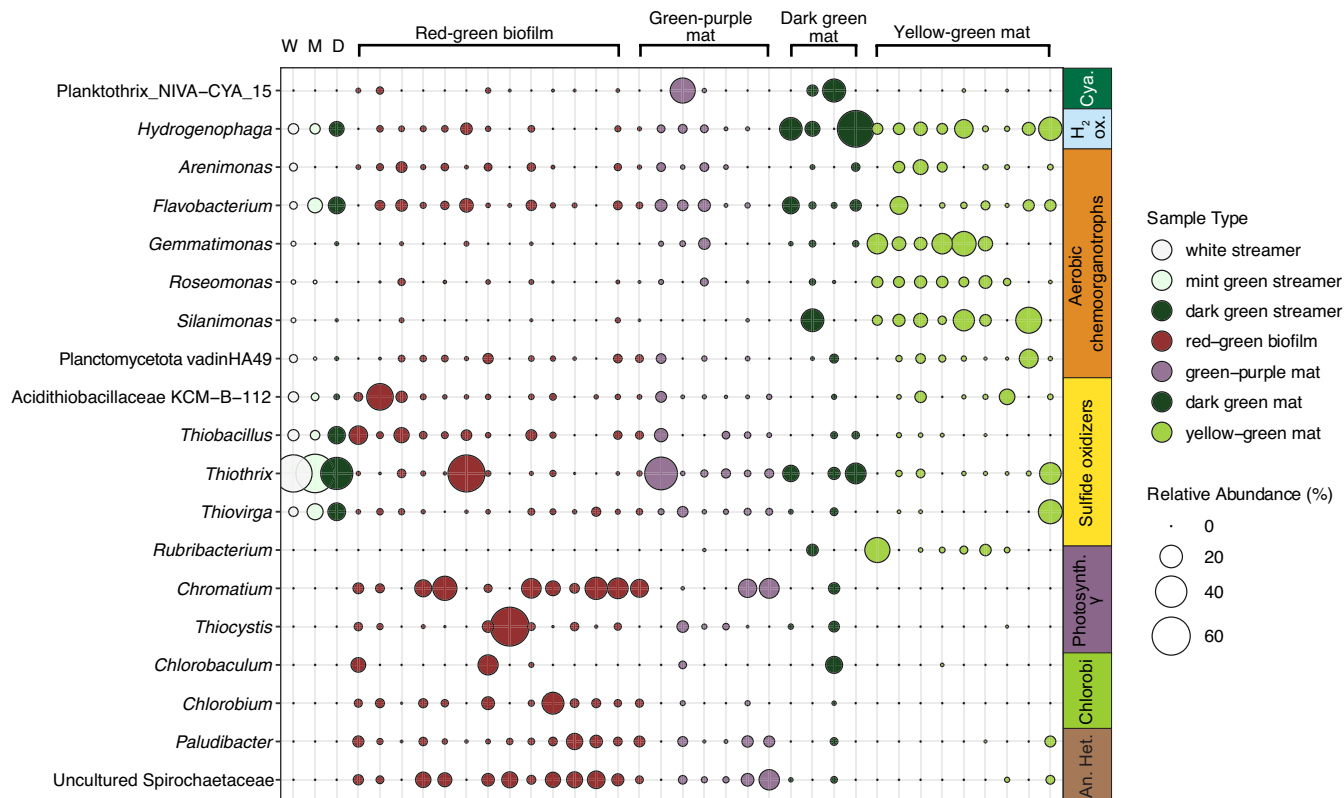


FIGURE 6 Biofilm and microbial mat microbial genera >1% total relative abundance. The x-axis represents individual samples. Representative filamentous streamer samples are also shown. “W” on the x-axis indicates a white streamer, “M” indicates a mint green streamer, and “D” indicates a dark green streamer. Taxonomic identification is based on 16S rRNA gene amplicon sequencing. Putative metabolic functions are based on metabolisms of the closest described relatives. Cy. = cyanobacteria, H₂ ox = hydrogen oxidizer, Photosynthetic γ = photosynthetic Gamma-proteobacteria, Chlorobi = photosynthetic Chlorobi, An. Het. = anaerobic chemoorganotroph

and biofilms in other freshwater environments (Figure 6). In contrast to the streamer communities, biofilms and microbial mats in slow-flowing pools had lower abundances of filamentous *Thiothrix* and were also colonized by anoxygenic photoautotrophs (Figure 6). These microbes, commonly described by their morphologies as purple and green sulfur/nonsulfur bacteria, use light as an energy source but use alternative electron donors such as H₂S or H₂. Anoxygenic photoautotrophs occupy a sunlit, anoxic niche and can provide a source of fixed carbon for heterotrophic microbes within the mats. Photosynthetic Gamma-proteobacteria of the genera *Chromatium*, *Thiocapsa*, and *Thiocystis* were present in red-green biofilms and purple-green mats, and likely contributed to the red and purple coloration of these biofacies. Under light and anoxic conditions and in the presence of reduced sulfur compounds (particularly sulfide), these photosynthetic Gamma-proteobacteria grow photolithoautotrophically and accumulate intracellular S⁰ globules (Dahl, 2008; Peduzzi et al., 2011). They are also capable of photoheterotrophic growth, and in the dark, are capable of chemolithoautotrophic growth using reduced sulfur as electron donors (Caumette et al., 1991; Dahl, 2008; De Wit & Van Gernerden, 1987; Peduzzi et al., 2011; Visscher & Taylor, 1993). These taxa were absent from yellow-green mats, which instead contained *Rubribacterium*—a genus of facultatively aerobic photoautotrophic and chemoorganotrophic Gamma-proteobacteria

(Boldareva et al., 2009). Photosynthetic Chlorobi of the genera *Chlorobium* and *Chlorobaculum* were abundant in red-green biofilms. Chlorobi are obligately anaerobic photolithoautotrophs that utilize reduced sulfur compounds (again particularly sulfide) as photosynthetic electron donors and deposit S⁰ extracellularly (Imhoff, 2014). Due to their tolerance for high sulfide concentrations (>2 mM), sensitivity to O₂, and adaptations to lower light intensities, they are typically found below purple photosynthetic Gamma-proteobacteria in a layered biofilm or mat (Imhoff, 2014). The S⁰ produced by chemolithotrophic and phototrophic sulfide oxidizers can be further oxidized to produce sulfate by some taxa found within SPC biofacies, including *Sulfurimonas*, *Sulfurovum*, *Thiomicrospira*, *Thiobacillus*, and *Thiovirga*. The mats were also colonized by hydrogen-oxidizing bacteria (e.g., *Hydrogenophaga*), and aerobic and anaerobic chemoorganotrophs (e.g., *Gemmatimonas*, *Silanimonas*, *Roseomonas*, *Paludibacter*, and uncultured Spirochaetaceae).

To measure the abundance and isotopic composition of sulfide within mats, silver strips were submerged in green-purple mats approximately 40m from the origin of the transect (Figure 1c, Pool 2) during illuminated daytime (mid-day) and simulated night conditions (Figure 7). During daylight conditions, captured sulfide was lowest at the mat surface and most abundant >1mm below the surface (Figure 7a). The deeper, more concentrated sulfide exhibited the

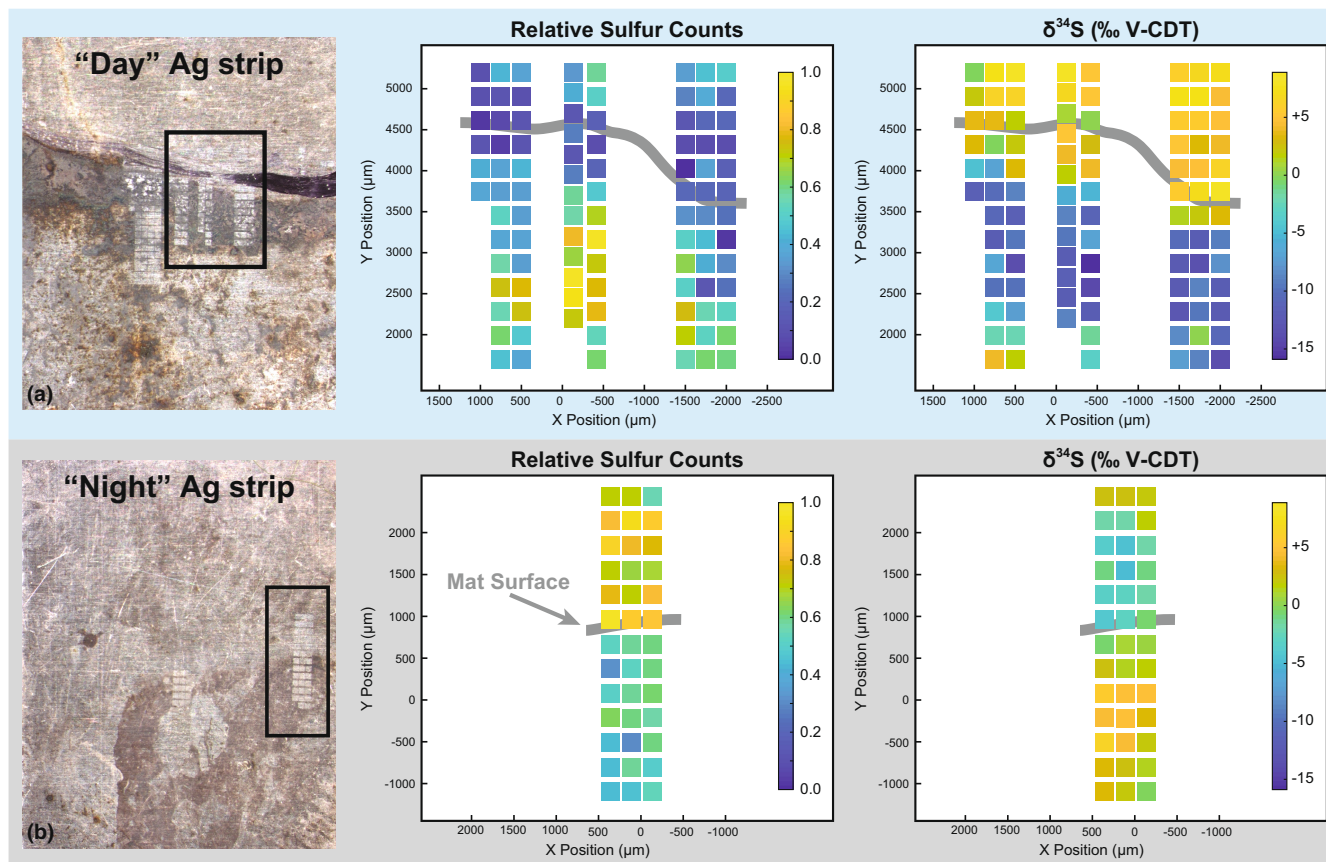


FIGURE 7 SIMS maps of sulfide abundance relative to each strip maximum and $\delta^{34}\text{S}$ values on silver strips submerged in green-purple mats. (a) Measurements from a silver strip submerged during illuminated daytime conditions; approximate mat surface position indicated by thick gray line. (b) Measurements from a silver strip submerged during simulated nighttime conditions

most depleted $\delta^{34}\text{S}$ values (−15‰ to −8‰), whereas sulfide at and above the mat surface was progressively enriched (0‰ to +8‰). $\delta^{34}\text{S}$ of sulfide below the surface of the mat was slightly more depleted than the $\delta^{34}\text{S}$ of sulfide in the bulk water (−13‰ to −2‰), and much more depleted than spring water sulfide (+4.6‰). In contrast, sulfide captured by the silver strip during simulated night conditions was most abundant at and above the mat surface, where $\delta^{34}\text{S}$ values of sulfide are moderately depleted (−5‰ to 0‰) compared to deeper in the mat (Figure 7b).

Within sediment samples ($n = 12$) *Desulfocapsa* was notably present at 1%–20% relative abundance (Figure S3). *Desulfocapsa* are strictly anaerobic Deltaproteobacteria that grow by the disproportionation of intermediate oxidation state sulfur species, including elemental sulfur, thiosulfate, and sulfite, to form sulfide and sulfate (Finster et al., 1998; Janssen et al., 1996). Also present within sediments were aerobic and facultatively anaerobic sulfur-oxidizing chemolithoautotrophs (e.g., *Thiovirga*, *Sulfurovum*, and *Thiobacillus*) and chemoorganotrophs (e.g., *Flavobacterium* and uncultured Anaerolineaceae). SPC bitumen samples ($n = 5$) were colonized by aerobic and anaerobic chemoorganotrophs, including *Tepidiphilus*,

Proteiniphilum, and uncultured Deferribacteraceae, that have been noted in other oil and petroleum-contaminated sites (Figure S3) (Tamazawa et al., 2017; Zhang et al., 2020).

Macrofauna at SPC, including frogs, tadpoles, and microbe-covered midge larvae and snails, were sampled to assess the microbial diversity of their associated microbiota. The skin of frogs ($n = 2$) and tadpoles ($n = 2$) was colonized primarily by *Burkholderia* spp., *Sphingomonas* spp., *Fusobacteria* spp., and *Desulfovibrio* spp. The microbial communities associated with midge larvae ($n = 2$) and snails ($n = 1$) were colonized by sulfide-oxidizing taxa, including *Thiothrix*, *Sulfurovum*, and *Sulfuricurvum*. Invertebrates in sulfide-rich environments often form symbioses with sulfide-oxidizing chemoautotrophic microorganisms. In these mutualistic nutritional symbiosis, microbial symbionts often detoxify sulfide and produce nutrients such as fixed carbon for the host. In return, the host provides the symbiont with optimal growth conditions, nutrients, and metabolites (Dattagupta et al., 2009; Dubilier et al., 2008; Stewart & Cavanaugh, 2006). It is possible that the sulfide-oxidizing microbial communities have a similar association with SPC macrofauna.

4 | DISCUSSION

4.1 | The microbially mediated sulfur cycle at Santa Paula Creek

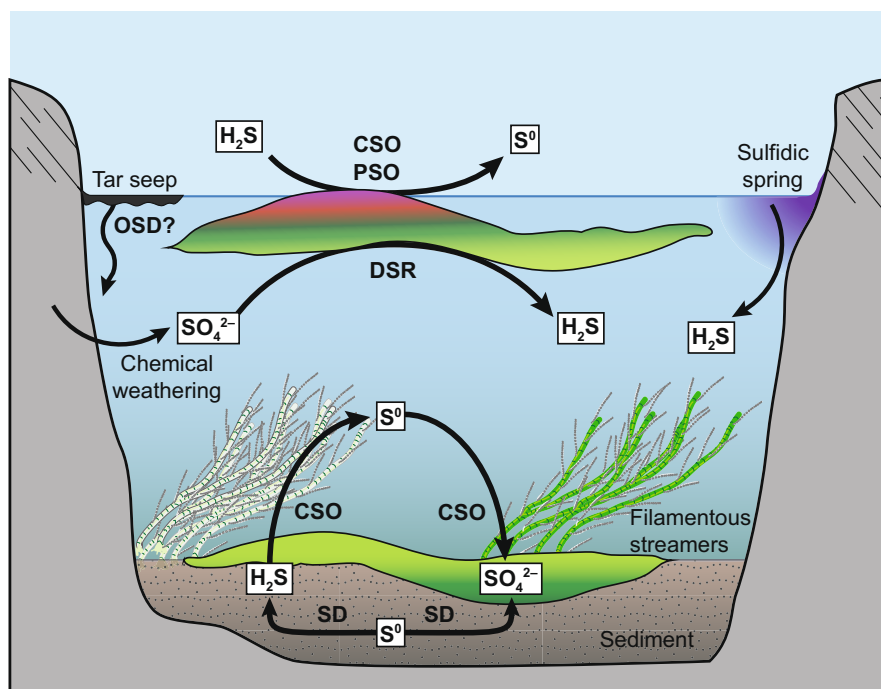
Microbial communities at SPC were colonized by taxa that are capable of diverse forms of sulfur metabolism, including the oxidation, reduction, and disproportionation of key sulfur-bearing species, thus providing an interesting mesocosm of players and processes in the biogeochemical sulfur cycle (Figures 6 and 8, Figure S2). At SPC, chemolithotrophic oxidation of sulfide to elemental sulfur serves as an exergonic metabolism, yielding approximately $198 \text{ kJ mol S}^{-1}$ under in situ conditions. All biofacies at SPC contained high abundances of microorganisms that are known to oxidize hydrogen sulfide and store it as S^0 , including *Thiothrix*, *Thiobacillus*, *Thiovirga*, *Sulfurovum*, and Acidithiobacillaceae KCM-B-112 (Figure 6, Figure S2). Sulfide oxidation may also be carried out within the microbial mats by anoxygenic phototrophic bacteria, which utilize sulfide rather than water as a reductant for photosynthesis and produce S^0 .

Bulk $\delta^{34}\text{S}$ values of white streamers were low relative to those of stream and spring water sulfate and sulfide, and were the most ^{34}S -depleted samples in the dataset. The significant isotopic fractionation and increasing $\delta^{34}\text{S}$ composition of white streamers along the sampling transect was likely driven by microbial activity. Microscale $\delta^{34}\text{S}$ trends within microbial mats indicated a diurnal production and vertical flux of depleted sulfide out of the mats (Figure 7). In the presence of light, anoxygenic phototrophic microbes within the microbial mats oxidized sulfide to produce S^0 , causing depleted S to remain below the surface of the mat. Without light, anoxygenic phototrophy could not occur and the decreased activity of phototrophs caused some of the depleted sulfide to flux vertically out of the mat. Although biological sulfide oxidation is likely the predominant

metabolism in SPC mats, it was long thought to produce no significant sulfur fractionation effect (Chambers & Trudinger, 1979; Kaplan & Rittenberg, 1964). However, recent evidence shows that phototrophic and chemolithotrophic sulfide oxidation may produce isotope fractionation, which could explain some of the fractionation measured at SPC (Pellerin et al., 2019; Zerkle et al., 2009). Similar experiments performed within other microbial mat systems attributed spatial variations of $\delta^{34}\text{S}_{\text{sulfide}}$ to the activity of bacterial sulfate reducers that were immediately adjacent to a layer of phototrophs (Fike et al., 2008). Within green-purple mats, dissimilatory sulfate-reducing and sulfur-disproportionating taxa, including *Desulfobulbus*, *Desulfomicrobium*, and *Desulfovibrio* were present from 1%–12% relative abundance. Sediments also contained high abundances of *Desulfocapsa* (1%–20% relative abundance), which grows by the disproportionation of intermediate oxidation state sulfur species (S^0 , thiosulfate, and sulfite) to form sulfide and sulfate (Figure S3). (Finster et al., 1998; Janssen et al., 1996). Microbial sulfate reduction and sulfur disproportionation cause large $\delta^{34}\text{S}$ depletions in sulfide of -4% to -46% (Canfield et al. 1998; Johnston et al. 2005; Canfield & Thamdrup, 1994; Chambers et al. 1975). The low $\delta^{34}\text{S}$ values in filamentous streamers and mats might also be explained by the accumulation of ^{34}S -depleted sulfur generated through repeated cycles of sulfate reduction, sulfide oxidation, and sulfur disproportionation along the stream transect, which can cause large ^{34}S depletions (Canfield & Thamdrup, 1994).

Bulk total sulfur $\delta^{34}\text{S}$ values from the Monterey Formation ranged between $+4\%$ and $+17\%$ (CDT), which correlated with the isotopic composition of the spring waters measured in this study ($\delta^{34}\text{S}_{\text{sulfide}} = +4.6\%$, $\delta^{34}\text{S}_{\text{sulfate}} = +25\%$) (Zaback et al., 1993). $\delta^{34}\text{S}$ of aqueous sulfate (-10% to 0%) and sulfide (-13% to $+2\%$) at SPC became increasingly more enriched along the transect, which can be explained by dilution and fractionation by microbial metabolism.

FIGURE 8 Proposed sulfur cycling in Santa Paula Creek. OSD, organic-sulfur degradation; CSO, chemolithotrophic sulfide/sulfur oxidation; PSO, phototrophic sulfide oxidation; DSR, dissimilatory sulfate reduction; SD = sulfur disproportionation



These data, along with the data collected in this study, suggest that the high flux of sulfur weathering from the Monterey black shales is capable of sustaining dynamic sulfur-metabolizing microbial communities, which drive significant isotopic fractionation within the streams. Taken in aggregate, these data reveal a system in which sulfur abundance and isotope composition evolves significantly along the sampling transect, and the distinct isotopic compositions of the various species suggest that sulfur-metabolizing microbial communities progressively fractionate the dissolved sulfate and sulfide pools to more and more enriched values with flow.

Distinctive geologic and geochemical factors at SPC support a complex biogeochemical sulfur cycle that is driven by microbially mediated sulfur redox transformations (Figure 8). Oxidative weathering and desulfurization of the underlying Monterey Formation black shale introduce high concentrations of sulfate and sulfide, respectively, into the oxygenated, sunlit stream, making SPC a unique freshwater environment in which to study sulfur cycling and underlying geologic controls on community structure. Within SPC biofacies, microbes exist which catalyze the oxidative, reductive, and disproportionative branches of the sulfur cycle (Figure 8). Inputs of sulfide from the Monterey Formation serve as an energy-rich electron donor for microbial communities. All biofacies contained chemolithotrophic sulfide oxidizers and mats and biofilms contained phototrophic sulfide oxidizers, with taxa that are known to produce and store S^0 through sulfide oxidation. After sulfide is oxidized, the S^0 produced can be further oxidized to sulfate. Dissimilatory sulfate reducers in the mats reduce stream sulfate or biogenic sulfate to produce sulfide that is depleted in ^{34}S . S^0 in the sediment, likely sourcing from the intracellular S^0 in streamers and mats, is disproportionated by *Desulfocapsa* to regenerate sulfate and sulfide that can be utilized by overlying mat communities. The microbially mediated cycling of sulfur through three major branches of the sulfur cycle (sulfide oxidation, sulfate reduction, and sulfur disproportionation) allow for the formation of dense, complex microbial communities that create and occupy the various geochemical niches at SPC.

4.2 | Niche differentiation at Santa Paula Creek

Our observations of the distinct biofacies and microbial communities in the ecosystem at SPC suggested that this site displays microbial niche differentiation in response to dynamic environmental conditions. Physical and geochemical parameters create numerous ecological niches that allow for the succession of complex, stratified microbial communities within biofilms, mats, and streamers (Madigan & Jung, 2009; Prieto-Barajas et al., 2018). In turn, these geochemical niches may be further defined through the metabolisms of the microbial communities, which colonize them. At SPC, the availability of numerous electron donors (sulfide and organic carbon) and acceptors (oxygen, sulfate, and nitrate), water flow, and light regimes create niches that are filled by morphologically and taxonomically distinct microbial communities with diverse metabolic functions. The predominant niches at SPC include fast flowing, oxic,

sulfidic waters, stagnant, sunlit, sulfidic waters, and stagnant, dark, oxic, non-sulfidic waters.

Filamentous streamers at SPC occupied a niche with fast water flow that was both oxic and sulfidic and thus were colonized primarily by a relatively low-diversity community of sulfide-oxidizing bacteria. *Thiothrix*, which was present at up to 78% relative abundance, was likely the main driver of sulfide oxidation within the streamers (Figure S2). Filament production by these microbes facilitates attachment to substrates, such as rocks or microeukaryotes (Figure 3e,j; Lackey et al., 1965; Uphof, 1927), and appears to allow *Thiothrix* to access turbulent waters where well oxygenated water mixes with sulfidic water. *Thiothrix* has been shown to occupy a similar niche in other sulfidic stream environments, where sulfide, oxygen, and water flow determined niche differentiation of *Thiothrix* and other sulfide-oxidizing populations (Larkin & Strohl, 1983; Macalady et al., 2008).

In contrast, microbial mats and biofilms in slow-flowing pools contained microbial communities that were significantly more diverse in terms of taxa (calculated by the Shannon and Simpson diversity indices) and metabolic functions (Figure 6, Figure S4). The high abundance of anaerobic microorganisms within the mats was supported by the low DO concentrations, which were lowest in the stagnant, sulfidic pools. Primary production within mats and biofilms was likely carried out by anoxygenic phototrophs (Chlorobi and photosynthetic Gamma-proteobacteria). Note that cyanobacteria, which carry out oxygenic photosynthesis as a major energy and fixed carbon source in mats, were generally absent from SPC communities. The niche of oxygenic phototrophy was likely also filled by photosynthetic microeukaryotes (e.g., *Amphora*, *Chlorella*, *Euglena*, *Hazenia*, *Melosira*), which were excluded from this dataset by removal of chloroplast and mitochondrial tag sequences (7.7% of total sequences). Microeukaryotes were noted by light microscopy (Figure 3) but were not analyzed further in the tag sequence data.

Microbial mats and biofilms also contained hydrogen oxidizers of the genus *Hydrogenophaga* and aerobic and anaerobic chemoorganotrophs, which likely consume carbon fixed by anoxygenic phototrophs and autotrophic sulfide-oxidizing bacteria. The low numbers of cyanobacteria, sulfur utilizing bacteria (chemolithotrophic and anoxygenic photolithotrophic sulfide oxidizers) and anaerobic chemoorganotrophs in yellow-green mats and presence of purple nonsulfur bacteria and aerobic chemoorganotrophs indicate that these mats occupied a niche within SPC that is relatively dark, oxic, and non-sulfidic. Conversely, the abundance of cyanobacteria and sulfide-oxidizing bacteria in red-green biofilms, green-purple mats, and dark green mats indicates that these biofacies inhabited illuminated environments that are possibly more stagnant or closer to sulfide springs.

Because of the highly dynamic nature of the stream environment, SPC may also be an ideal location to study metabolic flexibility in response to fluctuating environmental conditions. The aqueous chemistry of the creek waters and springs feeding into SPC suggest that microbial communities are unlikely to be nitrogen-limited, which can be common in other marine hydrocarbon-seep

or hydrocarbon-contaminated communities (Head et al., 2006). The increase of $\delta^{15}\text{N}_{\text{nitrate}}$ along the transect reflects a cumulative effect of nitrate assimilation by microbial communities, a process by which ^{14}N is preferentially taken up, leaving the dissolved pool ^{15}N -enriched. Microbial communities at SPC likely assimilate nitrate and ammonium, but may also utilize nitrate as an electron acceptor under oxygen-limiting conditions. In addition to the ability to respire aerobically, several taxa within SPC biofacies (e.g., *Thiothrix*, *Thiobacillus*, *Sulfurovum*, and *Hydrogenophaga*), may have the ability to couple anaerobic respiration of nitrate with the oxidation of reduced sulfur compounds or organic compounds (Bernardet & Bowman, 2006; Cai et al., 2011; Inagaki et al., 2004; Kämpfer et al., 2005; Schedel & Trüper, 1980; Trubitsyn et al., 2013, 2014; Willems et al., 1989). The ability to shift between metabolic functions (e.g., aerobic and anaerobic respiration or autotrophy and heterotrophy) may be advantageous for microbes in a rapidly changing chemical environment.

The presence of numerous physical and geochemical niches at SPC and the distinct and diverse microbial communities, which inhabit them further illustrates the unique nature of the sample site and makes it an interesting location for further geobiological studies. At SPC, variations in water flow, sulfide concentration, dissolved oxygen concentration, and sunlight levels create distinct niches that might be common to other environments where sulfur cycling occurs. Where steep and variable geochemical gradients occur, the partitioning of microbial communities and metabolic functions may also occur in order to exploit the energetic opportunities that exist. Perhaps the presence of these niches at SPC reflects microbial "rules" for sulfur cycling more broadly in other environments with active sulfur cycles. By incorporating metagenomics, transcriptomics, detailed geochemical analyses, and thermodynamic calculations into future studies, we may be able to better understand the ways that microbial communities occupy niches to take advantage of as many energy-yielding metabolisms as possible.

4.3 | Hydrocarbon seeps at SPC

The presence of active hydrocarbon seeps is a distinctive feature of SPC, and the inputs of degradable bitumen and sulfur in both organic and inorganic forms from these seeps into the stream may significantly stimulate activity in stream microbial communities (Orr, 1986; Werne et al., 2008; Zaback & Pratt, 1992). Synchrotron XANES confirmed the proposed organic nature of the sulfur source in the bitumen. Furthermore, the asphalt degradation profile indicates a transformation of the dominant organic-sulfur compound(s) towards a higher-energy, more oxidized form of organic sulfur as well as the beginning of evidence for sulfate in the most degraded bitumen samples. A different degradation path likely exists in anoxic environments where sulfate would not be expected to appear. Previous studies on the rate of hydrocarbon degradation have found that oxygen exposure is a major control on degradation rates (Atlas, 1981; Borden & Bedient, 1986; Hambrick et al., 1980; Mille et al., 1988). This is further evidenced by the elevated sulfide concentrations in

the spring water contrasting with elevated sulfate concentrations in the upstream portion of the sampled stream reach.

It is possible that the microbial communities within these seeps and stream sediments contribute to the degradation and desulfurization of these hydrocarbons. Natural asphalts and freshwater and marine sediments host microbial communities that link sulfur cycling, via the respiration of oxidized sulfur species, to organic matter mineralization via the oxidation of organic carbon and anaerobic oxidation of hydrocarbons (Jørgensen et al., 2019; Kim & Crowley, 2007). Given the interactions of sulfur and hydrocarbon-rich lithology in the stream environment, it is likely that the sediment and bitumen microbial communities at SPC are involved in similar elemental cycling. Future studies of carbon isotopes at SPC may reveal a difference in carbon isotope composition in consumers of Monterey hydrocarbons compared to phototrophs and consumers of phototrophs. In addition to the sulfide-oxidizing taxa mentioned previously, aerobic and anaerobic chemoorganotrophic microorganisms that have been noted in other oil-contaminated sediments and soils were present in SPC bitumen and sediments (Figure S3). These taxa included *Flavobacterium*, uncultured Anaerolineaceae (phylum Chloroflexi), uncultured Deferribacteraceae (phylum Deferribacteres) and Bacteroidetes vadinHA17, which are thought to be involved in degradation of n-alkanes and petroleum in certain oil reservoirs and sulfur-rich environments (Chaudhary et al., 2019; Chaudhary & Kim, 2017, 2018; Chen et al., 2017; Liang et al., 2015; Rahman et al., 2002; Sherry et al., 2013; Silva et al., 2013; Tamazawa et al., 2017; Zheng et al., 2018). Metagenomic sequencing and cultivation efforts can be used to explore the presence of hydrocarbon and organic-sulfur degradation pathways in organisms within SPC bitumen and sediment samples in future research efforts.

5 | CONCLUSIONS

Santa Paula Creek is a unique freshwater system where morphologically and functionally diverse biofacies form in association with naturally occurring hydrocarbon seeps and sulfur springs. The flowing, sunlit, and oxygenated nature of SPC, along with steady inputs of reduced species and hydrocarbons and local areas of anoxia set it apart from other terrestrial sulfur cycling sites. SPC represents an intriguing site for sulfur geochemistry in freshwater bodies, hosting elevated sulfate and sulfide concentrations, with sulfide concentrations exceeding $450\mu\text{M}$ in the springs. The presence of various electron donors/acceptors, light and dioxygen, and water flow regimes create distinct ecological niches that allow for the development of numerous distinct biofacies within a small stream reach. 16S rRNA amplicon sequencing and macro and microscale geochemistry were used to detail the dynamic biogeochemical sulfur cycle at SPC, which is largely driven by phototrophic and chemolithotrophic sulfide oxidation. Simulated day-night experiments further revealed that sulfur cycling and flux out of biofacies may be diurnally influenced. Transport processes and dilution of chemical species along the sampling transect create a diverse suite of geochemical conditions and

gradients that can be exploited by microbial communities. The dynamic nature of the SPC environment makes it an ideal site to study the interesting relationships between microbiology and geochemical cycling.

Given these initial findings, there is ample evidence to investigate intriguing geobiological questions at SPC in the future. Because of the numerous geochemical and ecological niches that exist, an interesting area for future research could include cultivation-based experiments coupled with metagenomic and metatranscriptomic studies to investigate the metabolic flexibility of microorganisms under fluctuating environmental conditions. Additionally, while this study focused mainly on the microbially mediated sulfur cycle, SPC is an ideal and easily accessible site to study the ways that life takes advantage of elevated sources of complex carbon and sulfur. Studying autochthonous hydrocarbon and organosulfur degrading microbial populations at SPC may also have possible applications to bioremediation research. Finally, SPC serves as a promising potential analog for terrestrial sulfur-metabolizing microbial communities arising in the Proterozoic or Archean. The wide-ranging sulfur isotopic compositions at SPC and their links to modern microbial metabolisms can be used as an analog for constraining sulfur isotope analyses from ancient systems (Canfield, 2005). This preliminary characterization of SPC has revealed a fascinating, and accessible, environment in which to study underlying geobiological controls on microbial community composition and its role in terrestrial sulfur cycling dynamics.

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DATA AVAILABILITY STATEMENT

The sequence data that support the findings of this study are openly available in the NCBI Sequence Read Archive at <https://www.ncbi.nlm.nih.gov/sra>, accession number PRJNA841681.

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